

# MMH 201 – Business Calculus: Midterm (1/14/2025)

Full name: ..... TROY ID No.: .....

Q1. Find any relative extrema of  $f(x) = x\sqrt{4 - x^2}$ . State where the function is increasing or decreasing and determine points of inflection.

Q2. Find the absolute maximum and minimum values of the function  $f(x) = \frac{x^2}{2x^3+1}$  over  $[0; +\infty)$ , if they exist.

Q3. The total cost and total revenue functions for a company are

$$C(x) = x^3 - 15x^2 + 63x + 230, R(x) = x(103 - 2x).$$

- a) Write down the equation for the profit function  $P(x)$ . Hence determine the level of output,  $x$ , for which profit is maximized.
- b) Find the marginal revenue  $MR(x)$  and the marginal cost  $MC(x)$ . Verify that  $MR = MC$  when the profit function has a maximum value.

Q4. A rectangular box with a square base and a cover is to have a volume of  $1250 \text{ cm}^3$ . If the cost per square cm for the bottom is \$3, for the top is \$2, and for the sides is \$2, what should the dimensions be in order to minimize the cost?